

The mean serum urate concentration (from uric acid) in a sample of urine was measured to be 64 mg/L. If a 2.0 mL aliquot of the urine sample is analyzed, what is the total mass of the serum urate present in the aliquot?

- ☐ A. 3.2×10^{-5} g [4%]
☒ B. 1.3×10^{-4} g [70%]
☐ C. 3.2×10^{-2} g [6%]
☐ D. 1.3×10^{-1} g [18%]

Correct

70% answered correctly

Explanation:

Aliquot volume 2.0 mL urine	×	Metric unit conversion $\frac{1 \text{ L}}{1000 \text{ mL}}$	×	Sample urate concentration $\frac{64 \text{ mg urate}}{1 \text{ L urine}}$	×	Metric unit conversion $\frac{1 \text{ g}}{1000 \text{ mg}}$	=	Mass of urate in the aliquot $1.3 \times 10^{-4} \text{ g urate}$
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Solution **concentrations** are sometimes expressed in terms of the **mass** of a given substance **per unit volume** of solution. When such a concentration is known, if an aliquot of known volume is taken from the solution, the mass of the substance contained in the aliquot can be calculated by multiplying the **aliquot volume** by the solution concentration provided that both values are expressed with respect to the **same units of volume**.

The concentration of urate in the urine sample was measured to be 64 mg/L, and the aliquot taken was 2.0 mL. To determine the mass of urate in the aliquot, one measurement must be converted to the same units of volume as the other. Converting the aliquot volume from mL to L gives:

$$2.0 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}}$$

Multiplying the converted aliquot volume by the given urate concentration yields the mass of urate contained in the aliquot (in milligrams):

$$0.0020 \text{ L} \times \frac{64 \text{ mg}}{1 \text{ L}}$$

Stating the mass of the urate in grams is accomplished by an additional metric conversion:

$$0.128 \text{ mg}$$

Rounding the answer to two significant figures and expressing it in scientific notation gives 1.3×10^{-4} g.

(Choice A) This value is obtained if the urate concentration is divided, instead of multiplied, by the aliquot volume.

(Choice C) This value results from doing *both* of two errors: not performing a unit conversion on either the aliquot volume or the urate mass, and dividing the urate concentration by the aliquot volume instead of multiplying.

(Choice D) This value is off by a factor of 1000 and is likely the result of not performing a unit conversion on either the aliquot volume or the urate mass.

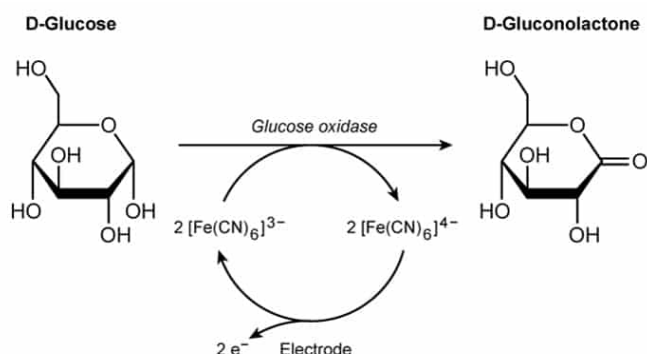
Educational objective:

Solution concentrations can be expressed as the mass of a given substance per unit of solution volume. The mass of a substance contained in an aliquot can be calculated by multiplying the aliquot volume by the solution concentration when both values are expressed with respect to the same units of volume.

Time spent: 0 sec
 Subject:
 General Chemistry

QID: 401777
 System:
 Solutions and Electrochemistry

Some test strips for blood glucose meters utilize $[\text{Fe}(\text{CN})_6]^{3-}$ as an electron transport mediator for the indirect electrochemical detection of the two-electron oxidation of D-glucose into D-gluconolactone, as shown in the figure below.



How many moles of electrons will be detected by the glucose meter if $160\text{ }\mu\text{g}$ of $[\text{Fe}(\text{CN})_6]^{3-}$ (molar mass 211.9 g/mol) react in a test strip during a measurement and each $[\text{Fe}(\text{CN})_6]^{3-}$ ion reacts only once?

- ☐ A. $0.38\text{ }\mu\text{mol}$ [8%]
- ✓ ☒ B. $0.76\text{ }\mu\text{mol}$ [60%]
- ☐ C. $1.32\text{ }\mu\text{mol}$ [11%]
- ☐ D. $1.51\text{ }\mu\text{mol}$ [19%]

Correct

60% answered correctly

Explanation:

Mass of mediator	Molar mass of mediator	Reaction mole ratio	Moles of electrons
$160\text{ }\mu\text{g } [\text{Fe}(\text{CN})_6]^{3-}$	$\frac{1\text{ }\mu\text{mol } [\text{Fe}(\text{CN})_6]^{3-}}{211.9\text{ }\mu\text{g } [\text{Fe}(\text{CN})_6]^{3-}}$	$\frac{2\text{ }\mu\text{mol } e^-}{2\text{ }\mu\text{mol } [\text{Fe}(\text{CN})_6]^{3-}}$	$0.76\text{ }\mu\text{mol } e^-$

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The **molar mass** of a compound is the amount of mass contained in 1 mole of the compound (grams per mole) and can be used to convert the **mass** of a compound to the number of **moles** of that compound. The **stoichiometry** (relative proportions) for the reactants and/or products of a particular reaction is given by the **mole ratios** from a **balanced reaction equation**. Stoichiometry can also be determined from an electrochemical scheme.

Given that $160\text{ }\mu\text{g}$ of $[\text{Fe}(\text{CN})_6]^{3-}$ react, the number of moles of $[\text{Fe}(\text{CN})_6]^{3-}$

contained in this mass is found by dividing by the molar mass of $[\text{Fe}(\text{CN})_6]^{3-}$ (scaled to micrograms):

$$\frac{211.9 \text{ g}}{\text{mol}} \times \frac{1 \mu\text{g}}{1 \times 10^{-6} \text{ g}} \times \frac{1 \times 10^{-6} \text{ mol}}{1 \mu\text{mol}} = \frac{211.9 \mu\text{g}}{\mu\text{mol}}$$

$$160 \mu\text{g} [\text{Fe}(\text{CN})_6]^{3-} \times \frac{1 \mu\text{mol} [\text{Fe}(\text{CN})_6]^{3-}}{211.9 \mu\text{g} [\text{Fe}(\text{CN})_6]^{3-}} = 0.76 \mu\text{mol} [\text{Fe}(\text{CN})_6]^{3-}$$

The stoichiometry in the electrochemical scheme shows that for every 2 electrons (e^-) conveyed to the electrode, the reduction of 2 $[\text{Fe}(\text{CN})_6]^{3-}$ ions is required. Therefore, if each $[\text{Fe}(\text{CN})_6]^{3-}$ ion reacts only once, the number of moles of $[\text{Fe}(\text{CN})_6]^{3-}$ that react is equivalent to the number of moles of electrons detected:

$$0.76 \mu\text{mol} [\text{Fe}(\text{CN})_6]^{3-} \times \frac{2 \mu\text{mol } e^-}{2 \mu\text{mol} [\text{Fe}(\text{CN})_6]^{3-}} = 0.76 \mu\text{mol } e^-$$

(Choice A) Using a 1:2 electron: $[\text{Fe}(\text{CN})_6]^{3-}$ mole ratio instead of the correct 2:2 ratio yields a result of 0.38 μmol .

(Choice C) A value of 1.32 μmol is obtained if the molar mass is mistakenly divided by the mass rather than dividing the mass by the molar mass.

(Choice D) A 2:1 electron: $[\text{Fe}(\text{CN})_6]^{3-}$ mole ratio used instead of the correct 2:2 ratio yields a result of 1.51 μmol .

Educational objective:

Given the mass of a compound sample, the number of moles present in the sample can be determined using its molar mass. Relating a known number of moles of a reactant to a balanced reaction equation, mole ratios can be applied to determine the stoichiometry of one chemical species relative to another species in a reaction.

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